

Problem 15.22

The accumulated value of an n year annuity-immediate is four times the present value of the same annuity. Calculate the accumulated value of 100 in $2n$ years.

Solution

For an annuity-immediate, the accumulated value and present value are related by

$$s_{\overline{n}|i} = a_{\overline{n}|i}(1+i)^n.$$

We are given that the accumulated value is four times the present value, so

$$s_{\overline{n}|i} = 4a_{\overline{n}|i}.$$

Substituting the relationship between $s_{\overline{n}|i}$ and $a_{\overline{n}|i}$ gives

$$a_{\overline{n}|i}(1+i)^n = 4a_{\overline{n}|i}.$$

Thus,

$$(1+i)^n = 4.$$

The accumulated value of 100 in $2n$ years is

$$100(1+i)^{2n}.$$

Since $(1+i)^n = 4$, we have

$$(1+i)^{2n} = ((1+i)^n)^2 = 4^2 = 16.$$

Therefore,

$$100(1+i)^{2n} = 100(16) = 1600.$$

Hence, the accumulated value is

$$\boxed{1600}.$$